UKAS QUALITY MANAGEMENT 005

SYSTEM CERTIFICATION ISO 9001:2008 SGS

ACROW

Formwork Technology

Acrow Bridge Table Form System

Acrow MISR Provides Full Site Service Based On More Than 35 Years Experience In The Field

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ACROW Formwork Technology

Introduction

Being one of the largest formwork and scaffolding companies in Egypt, we are introducing the bridge tables system. It is one of the most ideal systems to produce large concrete deck bridge areas, where the same deck bridge geometry is repeated several times.

In addition, all elements and accessories are assembled only once at the beginning of the job site, unlike other systems, where you have to re-assemble all the shoring components each time. Moreover, the elements could be easily moved as a whole movable unit that can be repositioned without wasting time on disassembly.

This is a quick installation and removal system which saves time and labor costs.

Acrow bridge table is one of the best solutions for any box girder bridge. Bridge deck system h20 & solider consist of panels connected together as per bridge deck shape. It is assembled in ground level and lifted as a one unit to its location. The shoring is either Acrow shorbrace system with high load capacity or ring system with it is modular components. shoring also can be moved as one unit horizontally by wheel jack unit or vertically by crane.

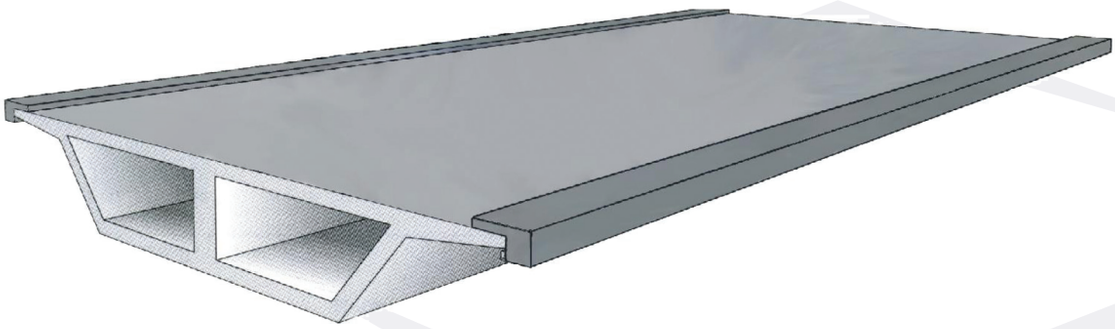
Transportation Wheel Jack Unite

HORIZONTAL SHIFTING WITH TRANSPORTATION WHEEL JACK UNIT

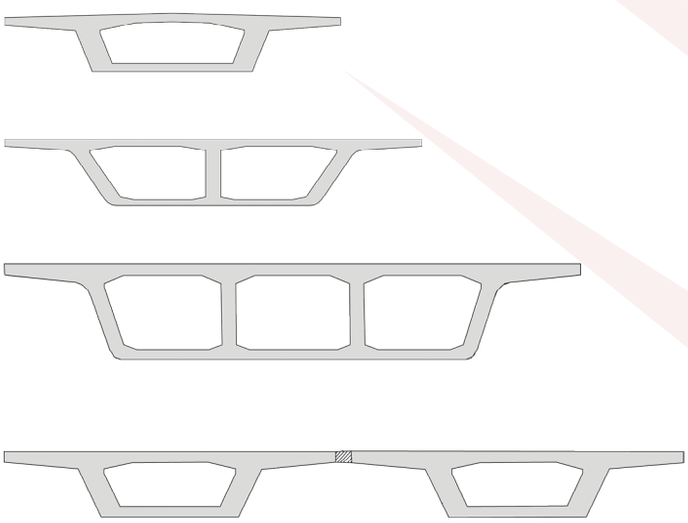
VERTICAL LIFTING BY CRANE

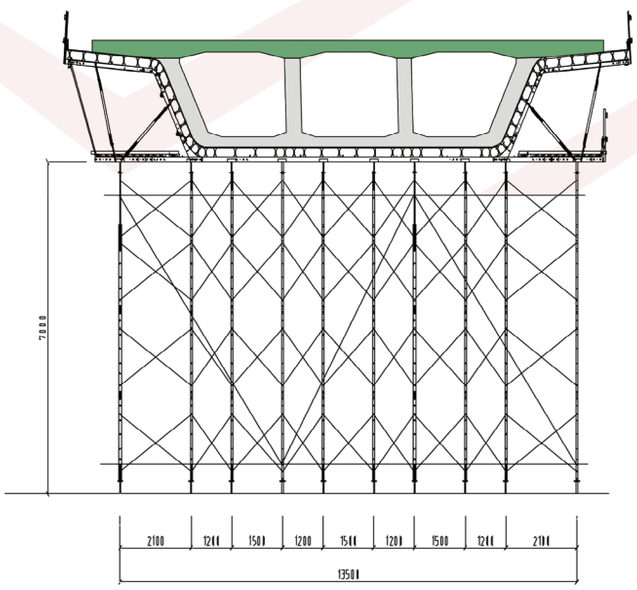
The highly efficient solution for any box girder deck bridge

- Types of cross sections for Box girder Bridge.



- cross section for double cell box girder bridge





System Components

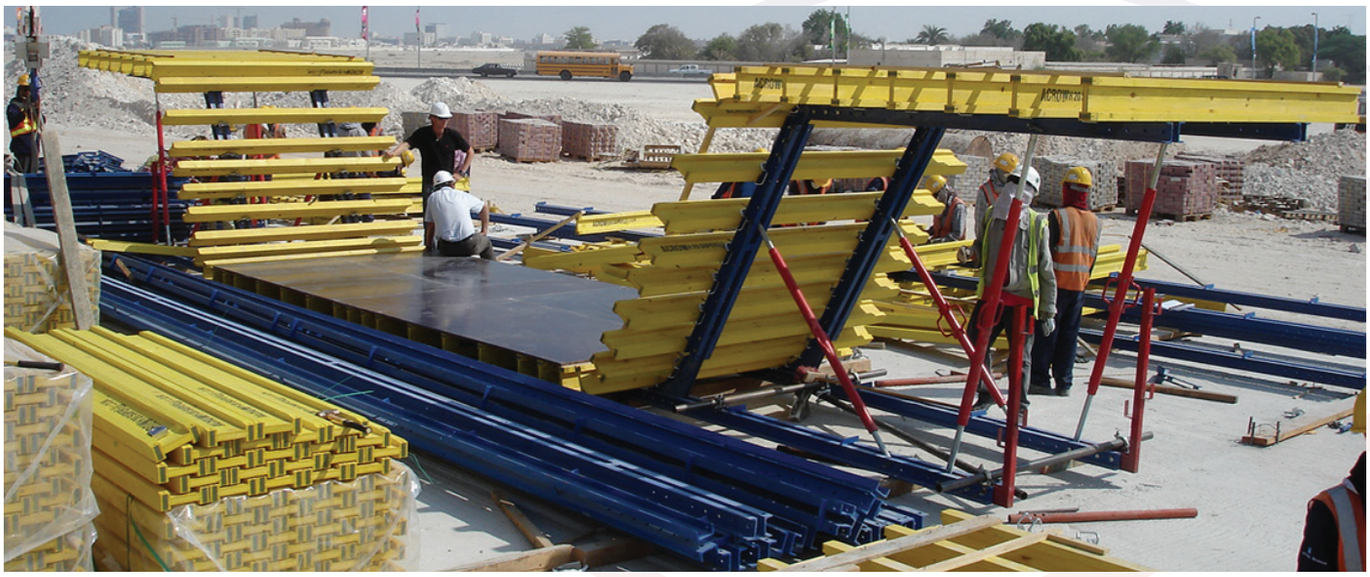
•**The shoring system:** is either tower which can be moved horizontally by means of wheel jack units, and vertically by crane. Moreover, it may be shorbrace frame with our standard frame 1.20*1.80m with or without telescopic frame at top of tower. In this case, there is only one frame size, It also ensures fast adjusting through the telescopic frame (52cm to 142.5cm) each 30cm increment. Lastly, Final adjustment could be reached through the (P.H.J) and (U.H.J.).

In addition, ring system can be used for shoring. In this case, the tower dimension is chosen according to the load and bridge geometry. Also, the required height can be achieved from the various modular vertical lengths. Finally, fine adjustment could be reached through (P.H.J) and (U.H.J).

System Components

•**Panels:** The whole decking consists of series of (H20 &Soldier) panels' independent to each other. The panel dimensions are chosen according to the shoring system and the bridge geometry. The length of panel is usually 2.5m for curved bridge zones and can be up to 5.00 m in case of straight bridge layout only.

•**Adjustment** of the external formwork is reached simply by using turn bukles (for external side formwork and for cantilever formwork).

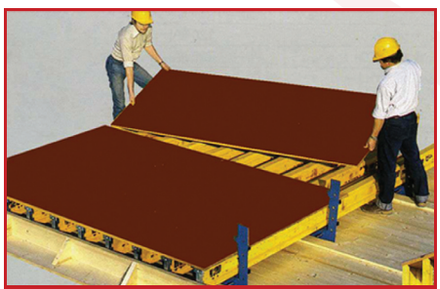


Panel assembly

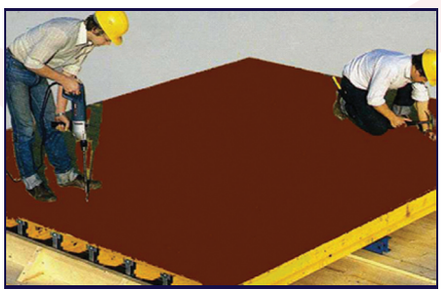
walers locations and timber girder spacing must be fixed on flat & leveled battern.

lay the H20 beam in their required positions

timber girder must be connected to the steel walers by H20 connection .



plywood must be fixed with the timber girder by torx screw .



holes for ties can drilled.

Panel Assembly

•The tables are assembled on the ground on a flat surface.
A plan is marked on the ground to aid in placing the Soldier and H20 timber beam.

•One the central part (bottom slab shuttering)has been assembled and the outer side shutter preassembly .
You can erect the side shutter and fix it with splices and adjust it by push pull props.

•Erection of the cantilever part and fixed with the side shutter by splices and adjustment the level by push pull prop.

Panel Assembly

Top Slab System

Shoring: The shoring may be special Acrow frame or sp. props according to the loads and the allowable decking opening.

Decking: The main Decking is H20 timber beam & the secondary is timber by site, which has the shape of the top slab ends.

Formwork Dismantling

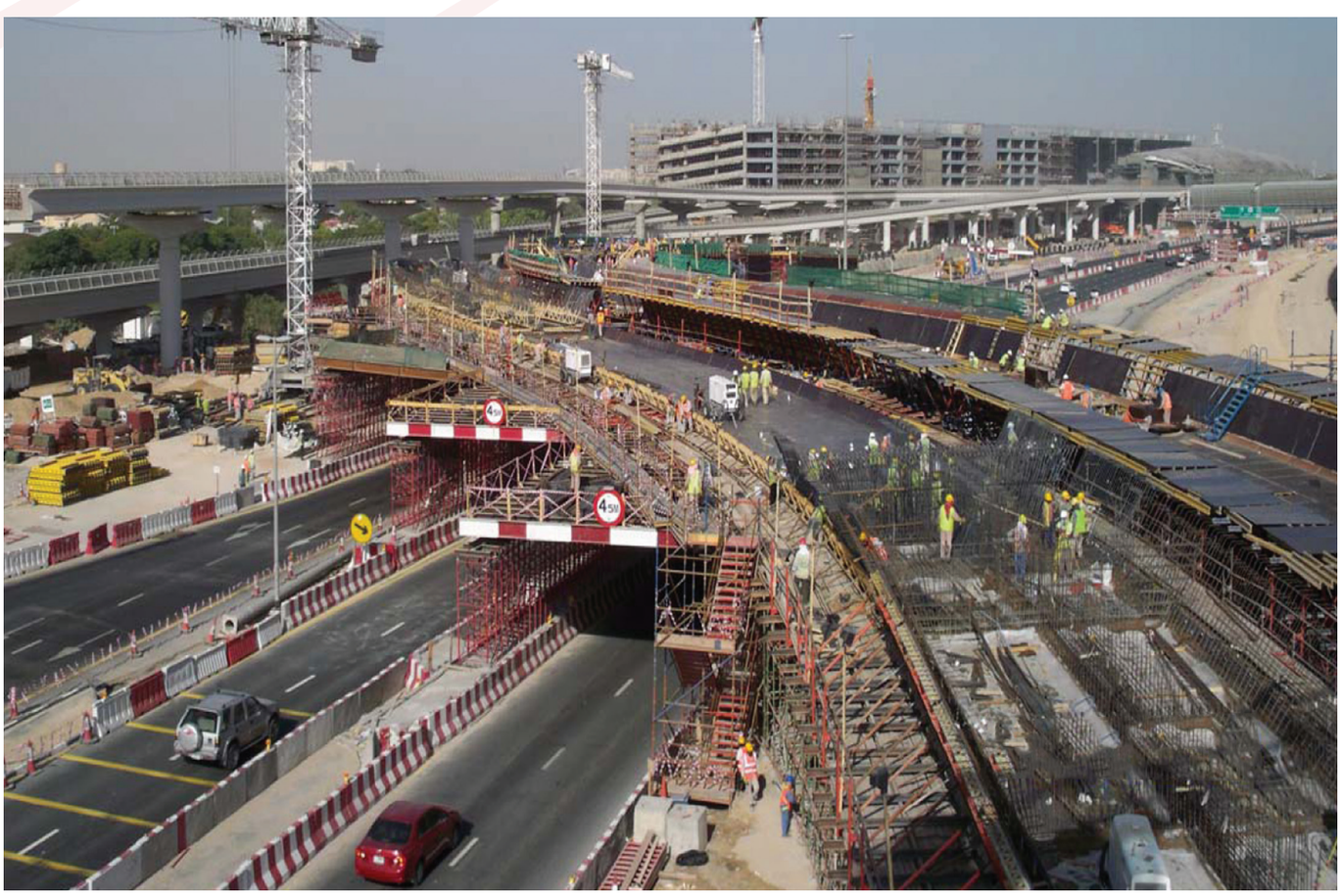
•Connect the external formwork to the bridge box girder using Acrow tie rod.
•Use Acrow Notched plate and Hex Nut instead on ordinary steel washer and wing nut.
•Lower the shoring and start moving it using Acrow transportation wheel jack unit.
•Connect the vertical tie to the lowering device machine using the Acrow chain rod coupler.
•Loose the Hex nut and remove the notched plate.
•Start the two Acrow lowering machines to lower the panels.
•Move the panel to its' new desired location using the trailer.
•Lift the panel to its' new location vertically using a crane.
• Acrow bridge table form is safe, fast, easy and economical.

Falsework Opening

•Technical Description: False work opening can be used during the construction in new bridges in case of any intersection with:

- Existing road.
- Small River.
- Another bridge.
- Railway

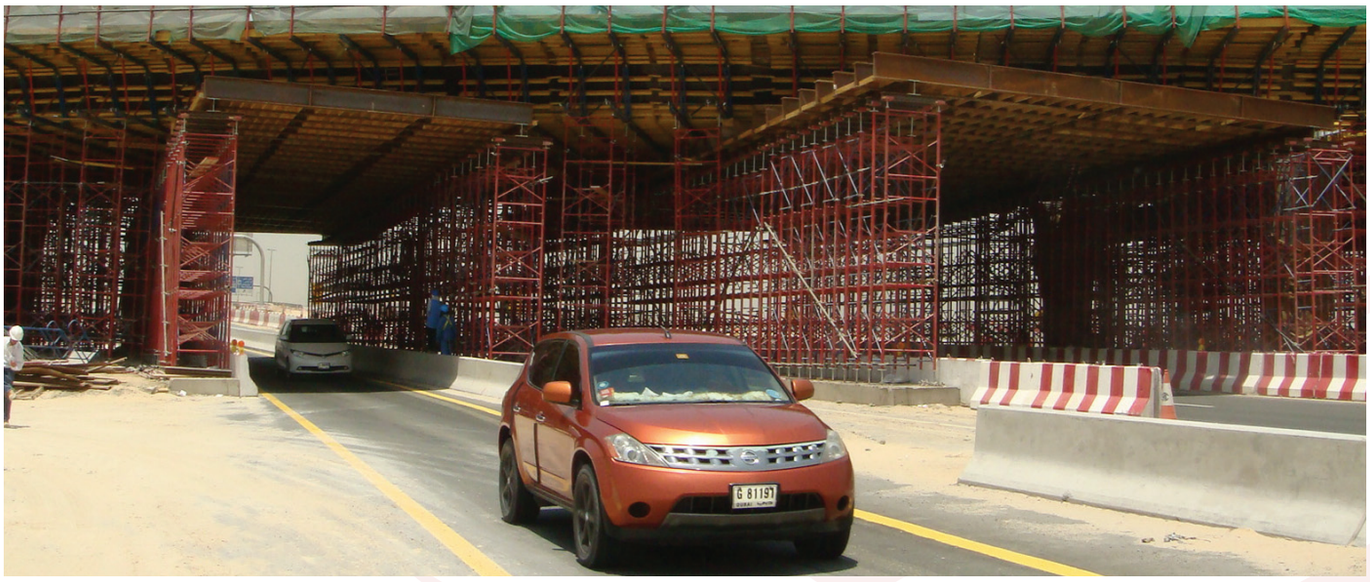
Therefore, by the application of false work opening, the loads of the bridge are transferred by the steel beams to the shoring and then directly to the soil.



Falsework Opening

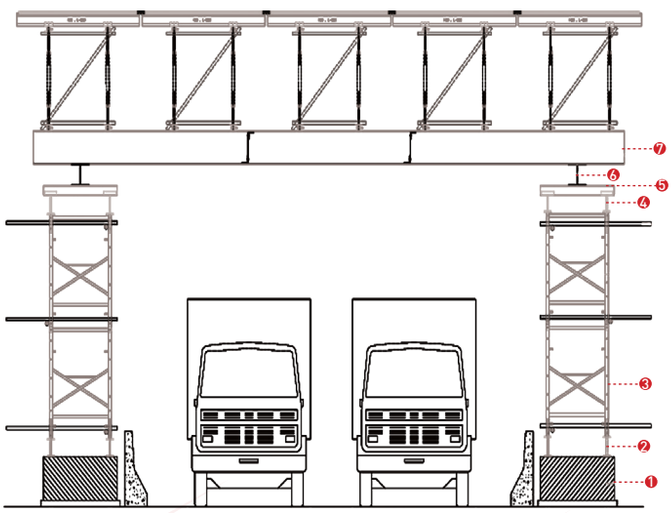
System advantage:

- Heavy Duty shoring system (Shorebrace) providing a maximum working load of 54.88 KN per leg.
- Simple and easily erected shoring system.
- Different heights can be applied for the system using the shorbrace frame.
- Standard and cheap steel beam decking which can be supplied to the site.
- Acrow bridge system can be used for both a one and two lane traffic where the maximum span doesn't exceed 12 m.
- Easily connection between the false work and the main bridge systems because of using the same system.
- Easily dismantling process.
- Safe, fast, easy and economical bridge system.



Falsework Opening

Erection system:
1-Concrete footing.
2-Post head jack.
3-Shorebrace frame.
4-U-Head jack.
5-Secondary Steel beam.
6-Main girder steel beam (1).
7-Main girder steel beam (2).

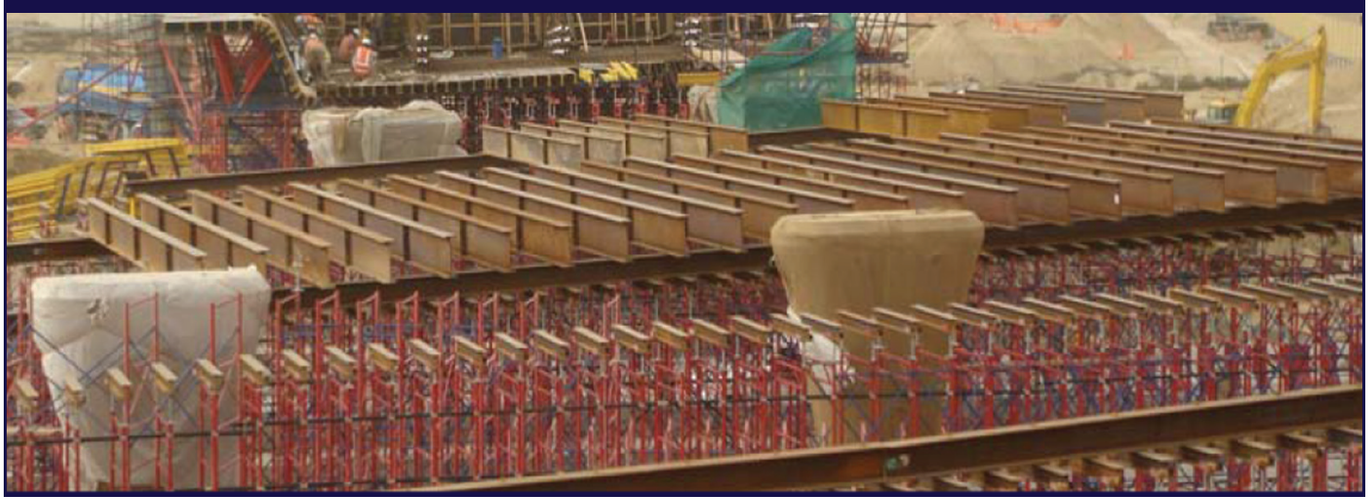


Falsework Opening For Tow Lane

Main Girder (1)

Shoring and Secondary beams

Main Girder (2)



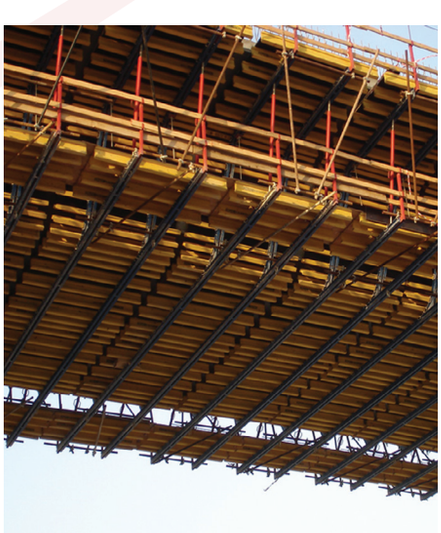
Falsework Opening With Acrow H.D.P

False work opening can be also done with Acrow heavy duty prop instead of shore brace system. Both steel beams and the foundation system will be designed according to loads and prop positions of the Acrow bridge system for both outer and inner decking. Finally, Tilt up shores may be used for some conditions.

Applications

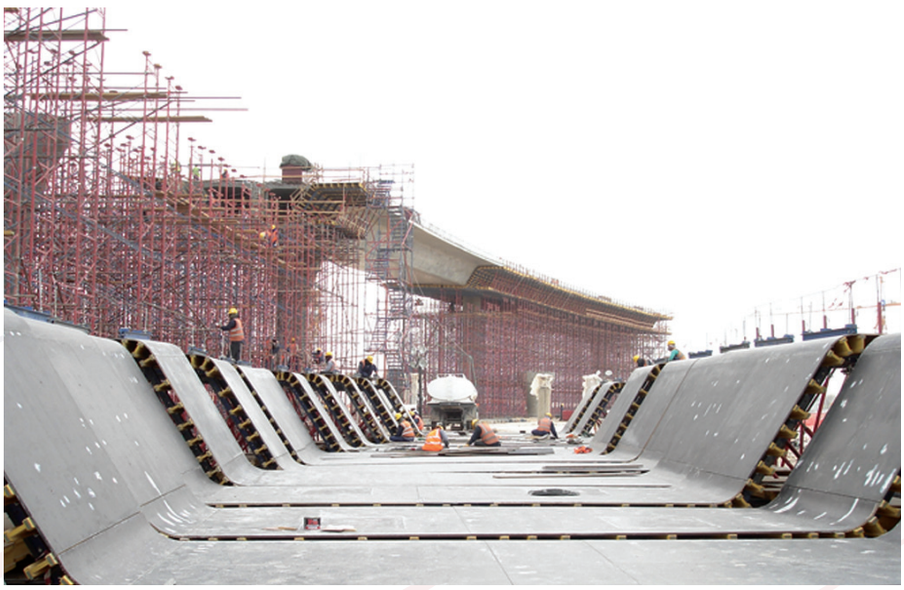
Application

Application

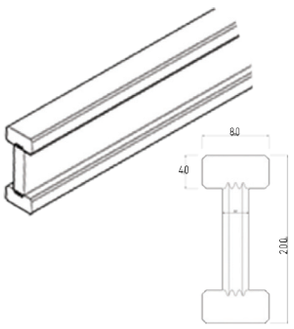
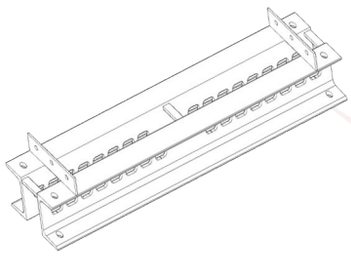
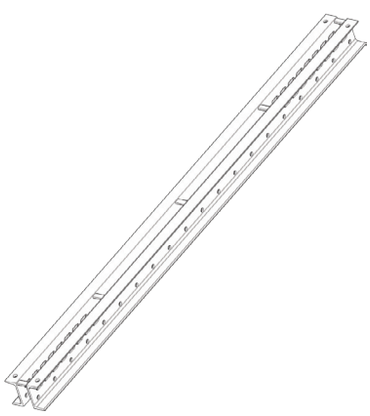


Application

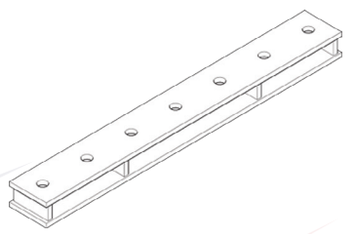
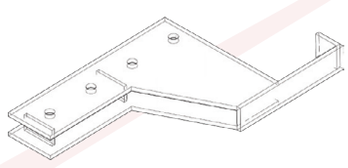
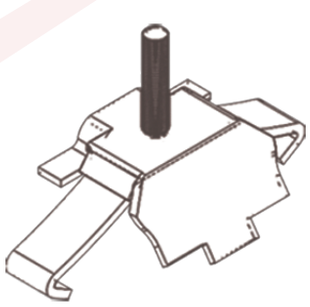
Application



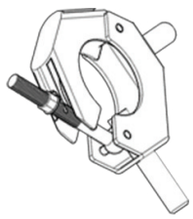
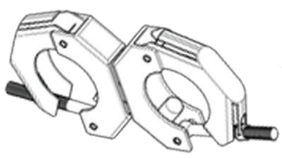
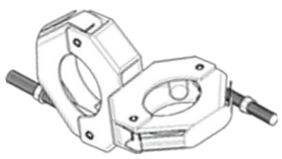
Components

Item	Description	Code no.	Weight (kg)
 Permissible bending Moment = 5.0 KNm Permissible shear force = 11.0 KN Approx. weight = 5.0 Kg/m	Timber H20 Beam		
	Timber H20 ,L= 2.50 M	6221177024062	12.50
	Timber H20 ,L= 2.90 M	6221177024079	14.50
	Timber H20 ,L= 3.30 M	6221177024093	16.50
	Timber H20 ,L= 3.60 M	6221177024109	18.00
	Timber H20 ,L= 3.90 M	6221177024116	19.50
	Timber H20 ,L= 4.50 M	6221177024130	22.50
	Timber H20 ,L= 4.90 M	6221177024147	24.50
	Timber H20 ,L= 5.90 M	6221177024154	29.50
	Soldier][10 (Holes)		
	Soldier][10 75 cm (Holes)	6221177020613	15.60
	Soldier][10 100 cm (Holes)	6221177020620	21.20
	Soldier][10 125 cm (Holes)	6221177020637	26.50
	Soldier][10 150 cm (Holes)	6221177020644	31.80
	Soldier][10 175 cm (Holes)	6221177020651	37.10
	Soldier][10 200 cm (Holes)	6221177020668	42.40
	Soldier][10 225 cm (Holes)	6221177020675	47.40
	Soldier][10 250 cm (Holes)	6221177020682	53.00
	Soldier][10 275 cm (Holes)	6221177020699	58.30
	Soldier][10 300 cm (Holes)	6221177020705	63.60
	Soldier][10 350 cm (Holes)	6221177020729	74.20
	Soldier][10 400 cm (Holes)	6221177020736	84.80
	Soldier][10 450 cm (Holes)	6221177020743	95.40
	Soldier][10 500 cm (Holes)	6221177020750	106.00
	Soldier][10 550 cm (Holes)	6221177020767	116.60
	Soldier][10 600 cm (Holes)	6221177020774	127.20
	Soldier][12 (slotted & Holes)		
	L= 1.17 m (slotted & Holes)	6221177020941	31.60
	L= 1.47 m (slotted & Holes)	6221177020958	39.70
	L= 1.78 m (slotted & Holes)	6221177020965	48.10
	L= 2.39 m (slotted & Holes)	6221177020972	64.50
	L= 3.00 m (slotted & Holes)	6221177033026	81.00
	L= 3.61 m (slotted & Holes)	6221177020989	97.50
	L= 4.22 m (slotted & Holes)	6221177020996	113.95
	L= 4.83 m (slotted & Holes)	6221177021009	130.45
	L= 5.44 m (slotted & Holes)	6221177021016	146.35
	L= 6.05 m (slotted & Holes)	6221177021023	163.35

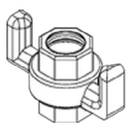
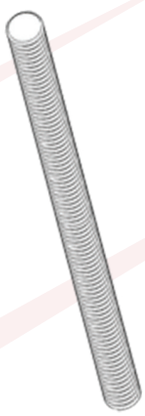
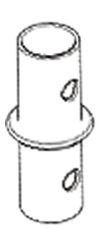
Components

Item	Description	Code no.	Weight (kg)
	splice 60 cm (for bridge)	6221177032555	7.00
	Diphram splice	6221177022105	7.10
	H20 Connection	6221177006891	0.70

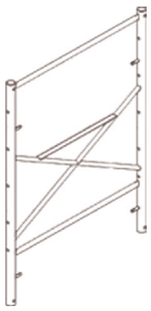
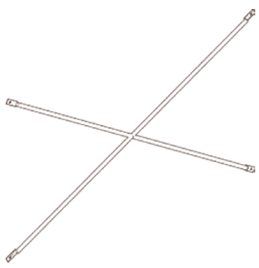
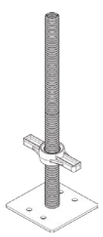
Components

Item	Description	Code no.	Weight (kg)
	Shorbrace Coupler	6221177007010	0.85
	pressed Prop double Coupler 1.5''x 2.0''	6221177006969	1.15
	presed Prop swivel Coupler 1.5''x 2.0''	6221177006976	1.25
	Scaffolding Tube ,L=1.00 m Scaffolding Tube,L=1.50 m Scaffolding Tube,L=2.00 m Scaffolding Tube,L=2.50 m	6221177017354 6221177017385 6221177017392 6221177017408	2.72 4.08 5.44 6.80

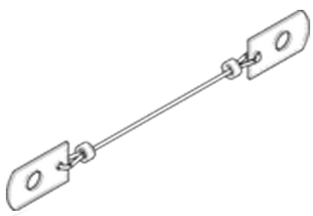
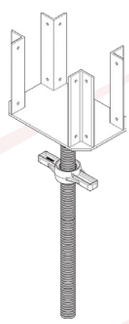
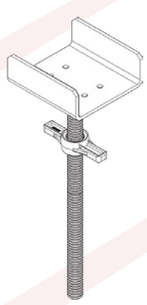
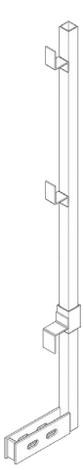
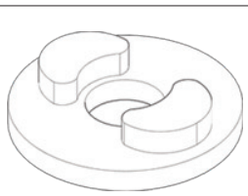
Components

Item	Description	Code no.	Weight (kg)
	Wing Nut DVD Wing Nut 19mm	6221177011383 6221177011390	0.45 0.55
	16mm diameter - safe working load 9.0 tons tension. L=0.75 mm L=1.00 mm L=1.25 mm L=1.50 mm L=1.75 mm L=2.00 mm L=2.50 mm L=3.00 mm L=3.50 mm L=4.00 mm L=5.00 mm L=6.00 mm Tie Rod Coupler DVD	 6221177023270 6221177023294 6221177023300 6221177023317 6221177023324 6221177023331 6221177023355 6221177023362 6221177023379 6221177023386 6221177023409 6221177023423 6221177007041	 1.12 1.50 1.86 2.24 2.61 2.98 3.73 4.48 5.22 5.97 7.46 8.95 0.25
	Steel Washer	6221177027513	1.10
	Shorebrace Frame Connector	6221177006341	0.50

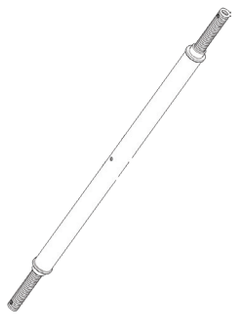
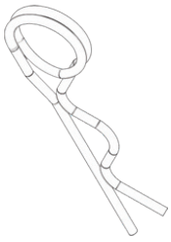
Components

Item	Description	Code no.	Weight (kg)
	Shorbrace frame 120 cm x 180 cm	6221177009359	29.0
	Shorbrace telescopic frame 120 cm * 165 cm	6221177009458	20.0
	CROSS BRACE (L x 1.20m) 0.90x1.20m 1.20x1.20m 1.50x1.20m 1.80x1.20m 2.10x1.20m 2.40x1.20m 2.70x1.20m 3.00x1.20m	 6221177008062 6221177008079 6221177008086 6221177008093 6221177008109 6221177008116 6221177008123 6221177008130	 3.35 3.80 4.70 5.20 5.80 6.40 7.15 7.70
	Post Head Jack Solid 34mm (L) 2”	6221177013110	7.10

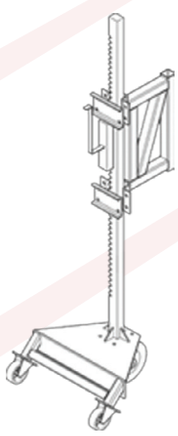
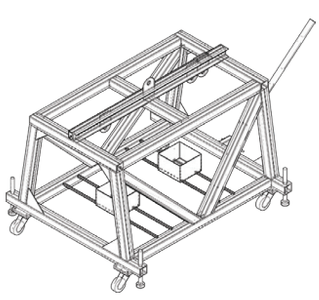
Components

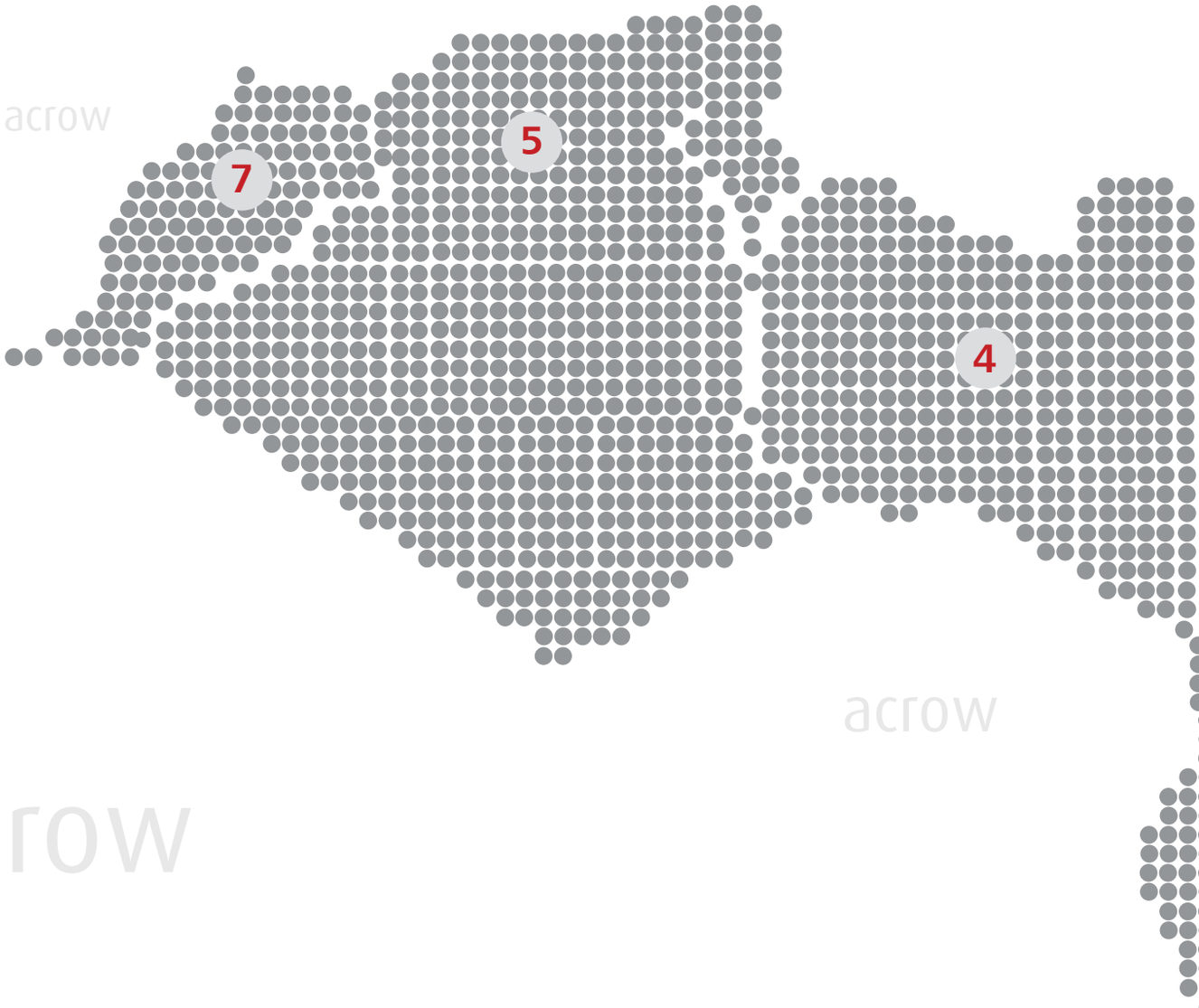
Item	Description	Code no.	Weight (kg)
	Safty Strap (Wire With 2-Plate)	6221177028282	0.18
	U-Head Jack Angle Type (For H20)	6221177025892	8.70
	U-Head Jack Solid-34mm, W=18 Cm , 2"	6221177025939	7.40
	HANDRAIL (for soldier) (slotted)	6221177032586	9.35
	COMPENSATION WASHER	6221177033200	0.126

Components

Item	Description	Code no.	weight(kg)
	Turn Buckle Push Pull PPS 131 Push Pull PPS 132 Push Pull PPS 161 Push Pull PPS 162 Push Pull PPS 201 Push Pull PPS 202 Push Pull PPS 252 Push Pull PPS 302 Push Pull PPS 352	 6221177014254 6221177014261 6221177014278 6221177014285 6221177014292 6221177014308 6221177014315 6221177014322 6221177014339	 9.00 11.40 9.83 12.43 13.14 14.00 15.70 19.00 23.10
	Big Wedge	6221177027629	0.85
	Rivet Pin 12.5mm Rivet Pin 7.50mm	6221177012298 6221177012281	0.20 0.12
	spring clip (large) spring clip (small)	6221177022617 6221177022624	0.03 0.01

Components

Item	Description	Code no.	weight(kg)
	bolt m 20x100	2012060027925	0.17
	nut m20	2012180029649	0.03
	Wheel Jack Unit transportation wheel jack (70 cm)	6221177027896	52.00
	Lowering Chair (W = 1.20M) (without motors)	6221177011239	705.00



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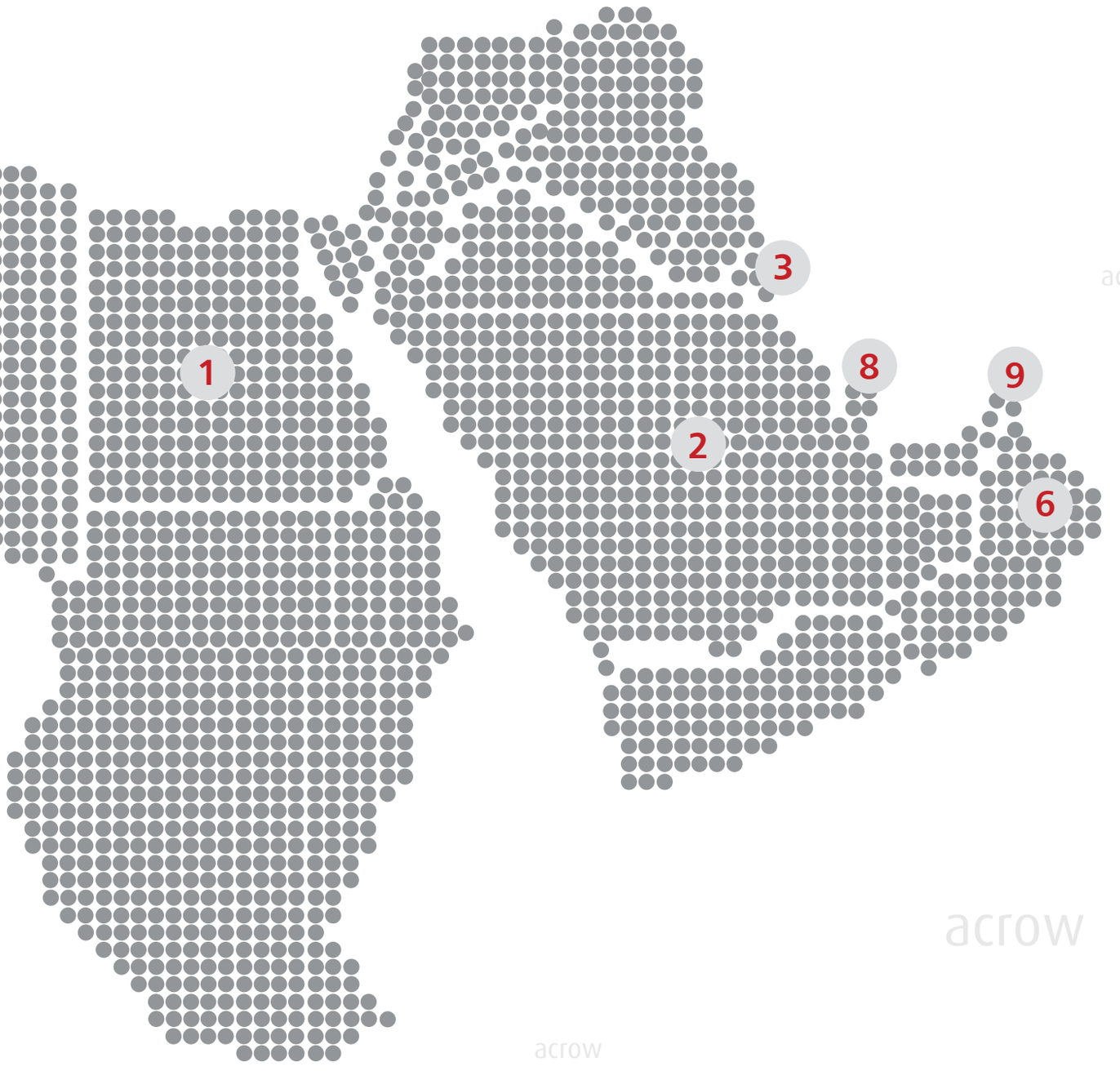

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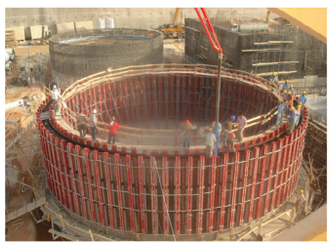
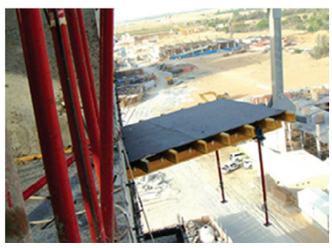
Acrow Big steel panel

Acrow European prop


Acrow Vari form


Acrow Prop table form

Acrow frame

Acrow early striking system

Acrow Ring lock

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